

DESIGN AND ANALYSIS OF ALGORITHMS

Practical No. 2: Quick Sort

1. Aim

To sort a given array in ascending order using the Quick Sort algorithm.

2. Objective

- Understand the divide-and-conquer strategy.
- Implement pivot selection and partitioning.
- Analyze the time and space complexity of Quick Sort.

3. Theory

Quick Sort is a comparison-based sorting algorithm based on divide and conquer. An element is selected as a pivot and the array is partitioned so that smaller elements are placed on one side and larger elements on the other. The process is recursively applied to both subarrays. Best and average time complexity is $O(n \log n)$, while the worst case is $O(n^2)$ for highly unbalanced partitions.

4. Conclusion

Quick Sort efficiently sorts an array by repeatedly partitioning it around a pivot. Its usual performance is $O(n \log n)$, although poor pivot selection can result in $O(n^2)$ time.